

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1.

The Gamma distribution is a conjugate prior for the mean of a normal distribution with known variance.

2 / 2 points

True

False

Nice work

Nice!

2.

True or False: The Beta distribution is a conjugate prior for the Bernoulli likelihood.

2 / 2 points

True

False

Nice work

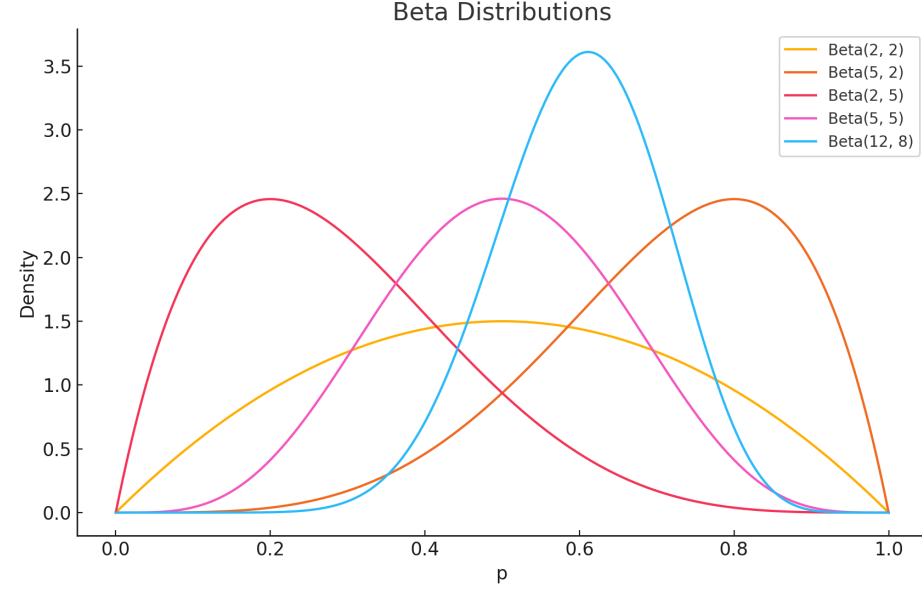
Correct!

3.

Which beta distribution best describes the following prior belief about a parameter p ?

2 / 2 points

“I am relatively uncertain, but I believe that the parameter is most likely to be near 0.5.”



Beta(12,8)

Beta(5,5)

Beta(2,2)

Beta(5,2)

Nice work

Nice!

4.

True or False: Consider the beta-beta conjugate relationship derived in this module. As the sample size (n) grows larger, the mean of the posterior will get closer to the maximum likelihood estimator.

2 / 2 points

True

False

Nice work

Nice!

5.

True or False: Consider the beta-beta conjugate relationship derived in this module. As the sample size (n) grows larger, the mean of the posterior will get closer to the mean of the data.

2 / 2 points

True

False

Nice work

Nice!

6.

True or False: Consider the beta-beta conjugate relationship derived in this module. As the sample size (n) grows larger, the mean of the posterior will get closer to the prior mean.

2 / 2 points

True

False

Nice work

Nice!

7.

Suppose researchers would like to know the probability, p , that an individual in a given population has a genetic marker that predisposes them for disease D (this genetic marker is such that an individual has it or doesn't; assume that our test for the marker is completely accurate). Researchers collected data in the following way: they tested people for this genetic marker until they found one person who had it. They stopped data collection at that point. Here's the data: $\{(0,0,1)\}$. Suppose your prior beliefs are best represented by a Beta(3,2) distribution. What is the posterior distribution for p ?

3 / 3 points

Beta(3,2)

Beta(5,5)

Beta(4,4)

None of the above.

Nice work

Nice!

8.

A company wants to estimate the average time customers spend on their website. They assume the time spent per customer follows an exponential distribution:
 $X | \lambda \sim \text{Exp}(\lambda)$
where λ is the rate parameter. They place a Gamma prior on λ :
 $\lambda \sim \text{Gamma}(\alpha, \beta)$
After observing n independent customer visit times $X_1, X_2, \dots, X_n$, what is the posterior distribution of λ ?

3 / 3 points

Gamma($\alpha + k, \beta$)

Gamma($\alpha + k, \beta + 1$)

Beta($\alpha + k, \beta + 1$)

Gamma($\alpha, \beta + k$)

Nice work

Nice! The posterior shape parameter is updated by adding k , and the rate parameter is updated by adding 1.

9.

Suppose you model the probability of a manufacturing defect using a Beta(5,5) prior. After inspecting 20 products, you find 8 defects. What is the expected posterior mean of p , the defect rate?

3 / 3 points

1/2

13/30

13/25

2/5

None of the above

Nice work

Nice!

10.

A researcher is studying the probability p of a website visitor clicking on an advertisement. They assume a Beta(2,3) prior for p . After observing 10 visitors, 4 clicked the ad. What is the posterior distribution for p ?

3 / 3 points

Beta(6,9)

Beta(4,6)

Beta(3,4)

Beta(6,7)

Nice work

Corect!